8.7.1 Predicting the extent of a reaction

The value of equilibrium constant, K_c tells us the extent of a reaction, i.e., it indicates how far the reaction has proceeded towards product formation at a given temperature.

A large value of K_c indicates that the reaction reaches equilibrium with high product yield. On the other hand, a low value of K_c indicates that the reaction reaches equilibrium with low product yield. In general, if the K_c is greater than the 10^3, the reaction proceeds nearly to completion. If it is less than 10^{-3}, the reaction rarely proceeds. If K_c is in the range 10^{-3} to10^3, significant amount of both reactants and products are present at equilibrium.

Table 8.2 Dependence of extent of reaction on K_C

Value of K_C	**$K_C < 10^{-3}$**	**$10^{-3} < K_C < 10^3$**	**$K_C > 10^3$**
Relative concentrations of reactants and products	[Products] << [Reactants]	Significant quantity of Products and Reactants	[Products] >> [Reactants]
Extent of reaction	Reaction makes a little progress in the forward direction	Both the forward and backward reactions make significant progress	Reaction nearly goes to completion
Prediction	Reverse reaction is favoured	Neither forward nor reverse reaction predominates	Forward reaction is favoured
Examples	**Decomposition of water at 500 K** $2H_2O(g) \rightleftharpoons 2H_2(g)+O_2(g)$ $K_C = 4.1 \times 10^{-48}$ **Oxidation of nitrogen at 1000 K** $N_2(g)+O_2(g) \rightleftharpoons 2NO(g)$ $K_C = 1\times 10^{-30}$	**Dissociation of bromine monochloride at 1000 K** $2BrCl(g) \rightleftharpoons Br_2(g)+Cl_2(g)$ $K_C = 5$ **Formation of HI at 700K** $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ $K_C = 57.0$	**Formation of HCl at 300K** $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$ $K_C = 4 \times 10^{31}$ **Oxidation of carbon monoxide at 1000 K** $2CO(g) + O_2(g) \rightleftharpoons 2CO_2(g)$ $K_C = 2.2 \times 10^{22}$

Example

Consider the following equilibrium reactions and relate their equilibrium constants

i) $N_2 + O_2 \rightleftharpoons 2NO$; K_1
ii) $2NO + O_2 \rightleftharpoons 2NO_2$; K_2
iii) $N_2 + 2O_2 \rightleftharpoons 2NO_2$; K_3

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